WG 1

Intelligent Systems

Exercise 1a: Given A = {1, 3, 8, 10} and B = {6, 9, 12} for a Universe U={1,3,6,8,9,10,12}. Which of the following statements is/are TRUE?

A) B is the complement of A
B) $A \cap B = \varnothing$
C) A and B are disjoint sets
D) B $\subset$A

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Exercise 1b: There are 25 elements in a universal set. If $|A| = 10$, $|B| = 10$ and $|A \cap B| = 10$, what is the cardinality A union B, formally $|A \cup B|$?

A) 0
B) 10
C) 15
D) 20

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Exercise 1b: There are 25 elements in a universal set. If $|A| = 10$, $|B| = 10$ and $|A \cap B| = 10$, what is the cardinality A union B, formally $|A \cup B|$?

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1. First determine all the possible combination of nodes
2. Determine all the possible graphs
3. Calculate with the formula how many can be: 2 to the power of number of possible combinations

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G={V,E} where the fully connected graph would have:

E = {{a,a},{a,b},{a,g},{b,b},{b,g},{g,g}}

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For each subset of E there is a different graph, i.e. E1 = the empty graph, E2={{a,a}}, E3={{a,a},{a,b}}, E4={{a,a}{a,g}}, etc. In total there are 2^6 = 64 different graphs.

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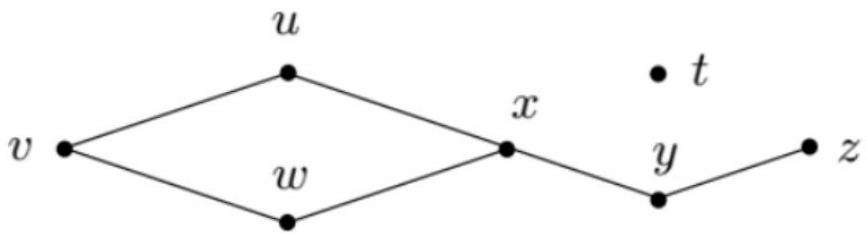
For a directed graph the fully connected graph would have

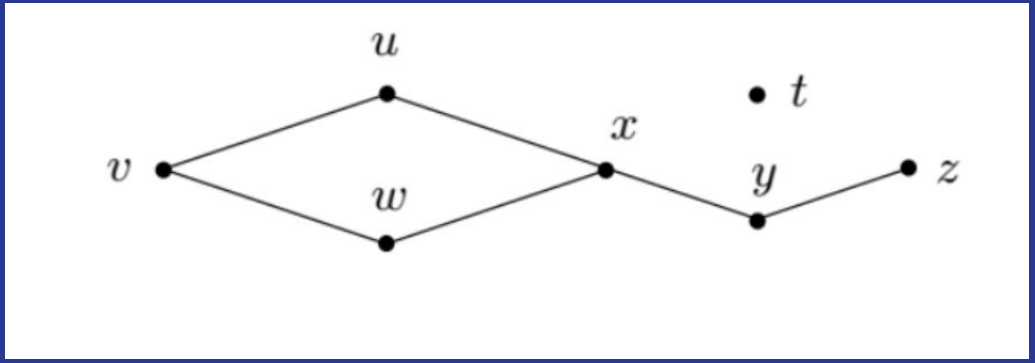
E={(a,a),(a,b),(b,b),(b,a)}, so in total 2^4=16 different graphs.

Exercise 2b: The adjacency matrix of a graph G is the matrix A defined as follows: adjacency for any two vertices u and v,

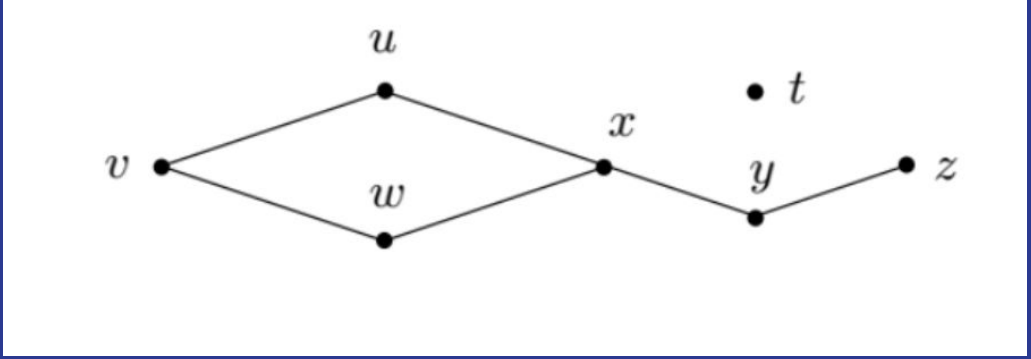
$$A[u,v] = \begin{array}{ll} 1 & \text{if } uv \in E_G\,, \\ 0 & \text{otherwise.} \end{array}$$

In other words, an adjacency matrix A for a graph G has a 1 if, and only if, there is an edge between nodes u and v, and a 0 otherwise. Write down the adjacency matrix for the graph:

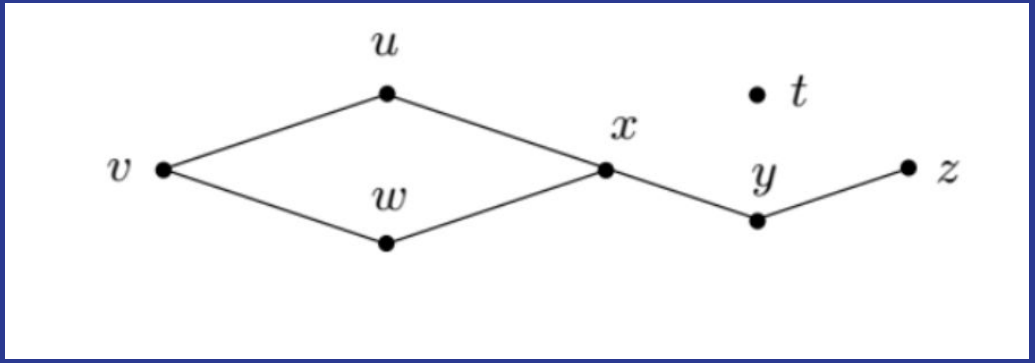




	V	U	W	X	T	Y	Z
V							
U							
W							
X							
T							
Y							
Z							



	V	U	W	X	T	Y	Z
V	0						
U							
W							
X							
T							
Y							
Z							



	V	U	W	X	T	Y	Z
V	0	1					
U							
W							
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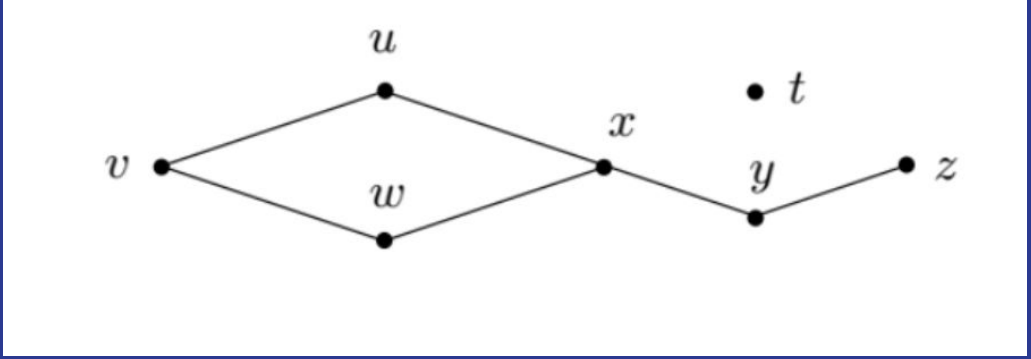
T							
Y							
Z							



	V	U	W	X	T	Y	Z
V	0	1	1				
U							
W							
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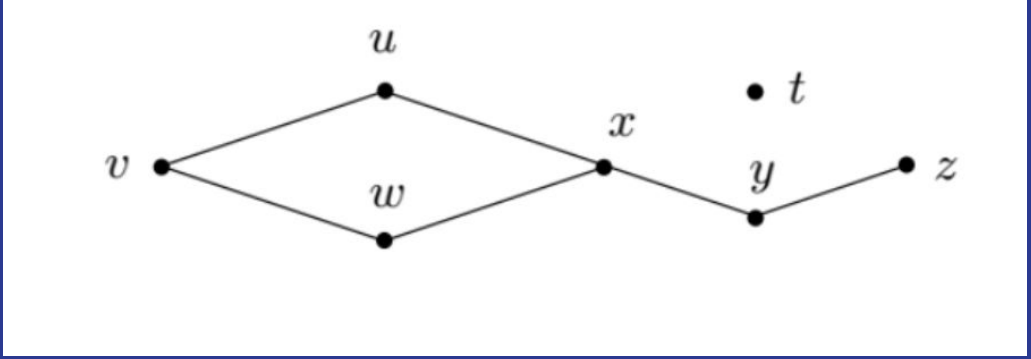

	V	U	W	X	T	Y	Z
V	0	1	1	0	0	0	0
U							
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X							
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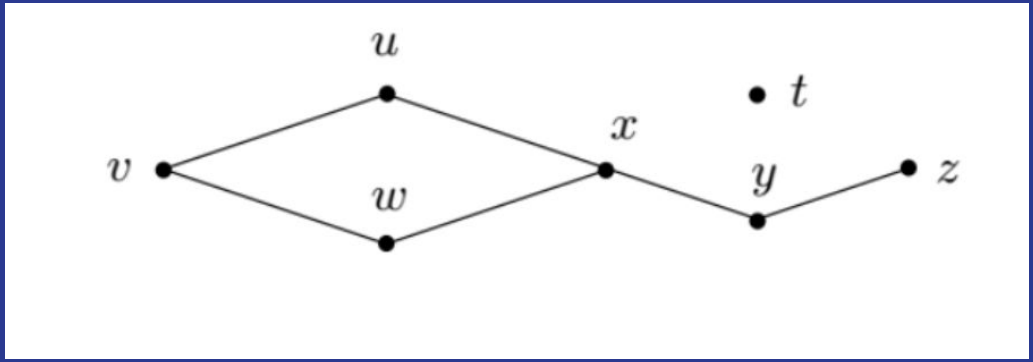
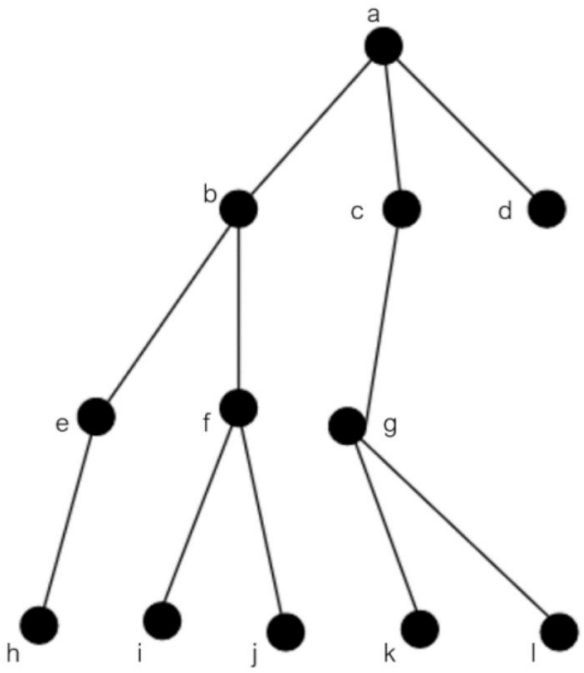


	v	u	w	x	t	y	z
v	0	1	1	0	0	0	0
u	1	0	0	1	0	0	0
w	1	0	0	1	0	0	0
x	0	1	1	0	0	1	0
t	0	0	0	0	0	0	0
y	0	0	0	1	0	0	1
z	0	0	0	0	0	1	0

Exercise 3: Annotate the following tree with labels: root and the leaves.

- What is the depths of the leaf-nodes, and thus the heights/depths of the tree?
- Which are the nodes with the same descendants as children?
- Which are the nodes which can be identified uniquely as the child of their parent?



Exercise 3: Annotate the following tree with labels: root and the leaves.

- What is the depths of the leaf-nodes, and thus the heights/depths of the tree? 3
- Which are the nodes with the same descendants as children?
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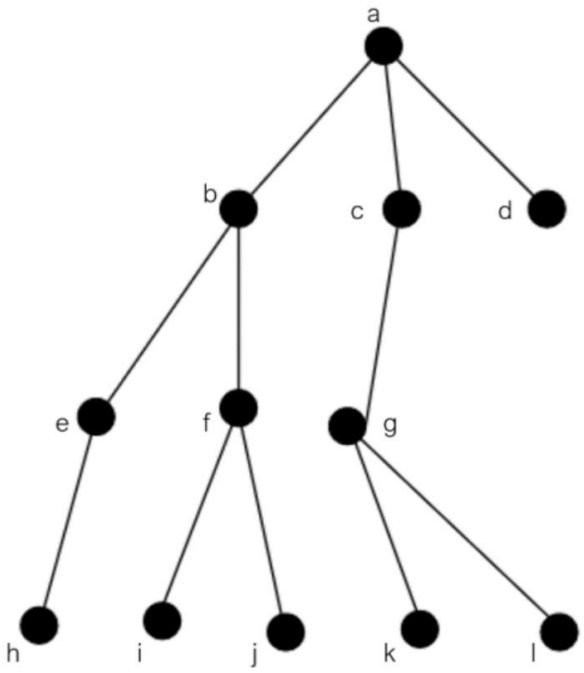

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- What is the depths of the leaf-nodes, and thus the heights/depths of the tree? 3
- Which are the nodes with the same descendants as children? D,E,F,G,H,I,J,K,L
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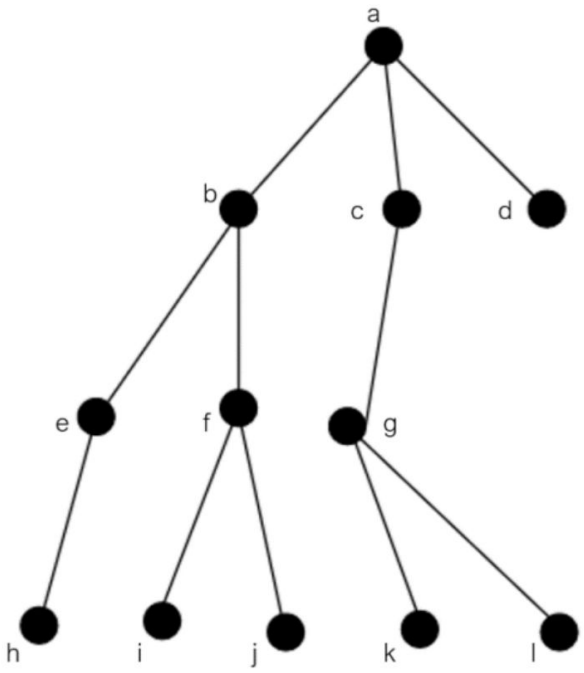

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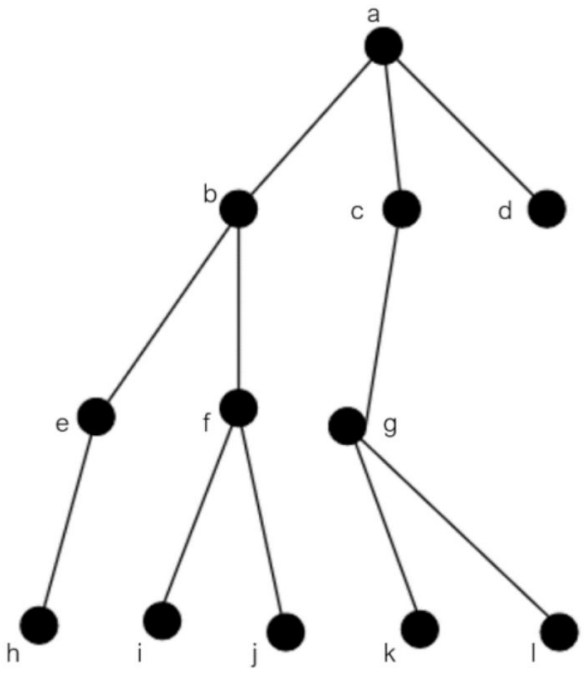
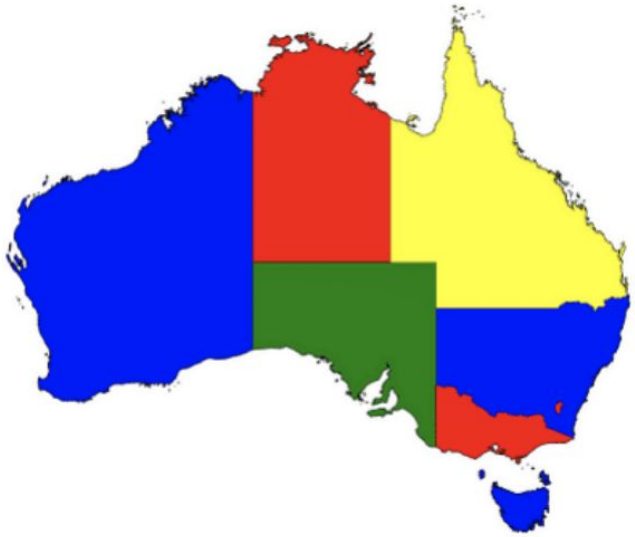


Exercise 4: Give the initial state, goal test, successor function and cost function for the following problem. Choose a formulation that is precise enough to be implemented.

- You have to color the planar map of Australia using only four colors in such a way that no two adjacent regions have the same color.



Initial state	= state that the agent starts in
goal test	= test which determines whether a given state is a goal state
successor function	= agent's options: given a state, it returns a set of <action, state/successor> pairs, where each state is the state reachable by taking the action. For example, from the state In(Arad), the successor function for the Romania problem would return: {<Go(Sibiu), In(Sibiu)>, <Go(Timisoara), In(Timisoara)>, <Go(Zerind), In(Zerind)>}
cost function	= a mathematical formula used to predict the cost associated with a certain action or a certain level of output

- You have to color the planar map of Australia using only four colors in such a way that no two adjacent regions have the same color.

Initial state:

Goal test:

Successor function:

Cost Function:

- You have to color the planar map of Australia using only four colors in such a way that no two adjacent regions have the same color.

Initial state:No region is colored

Goal test:

Successor function:

Cost Function:

- You have to color the planar map of Australia using only four colors in such a way that no two adjacent regions have the same color.

Initial state:No region is colored

Goal test: All colored + no adjacent

Successor function:

Cost Function:

- You have to color the planar map of Australia using only four colors in such a way that no two adjacent regions have the same color.

Initial state:No region is colored

Goal test: All colored + no adjacent

Successor function: Assign a color to a region

Cost Function:

- You have to color the planar map of Australia using only four colors in such a way that no two adjacent regions have the same color.

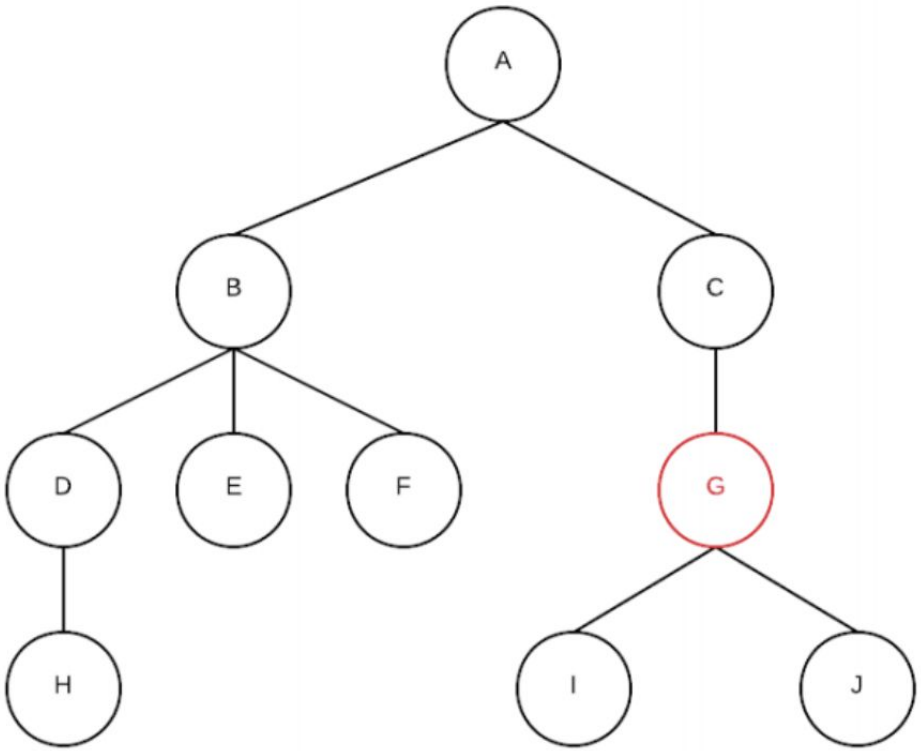
Initial state:No region is colored

Goal test: All colored + no adjacent

Successor function: Assign a color to a region

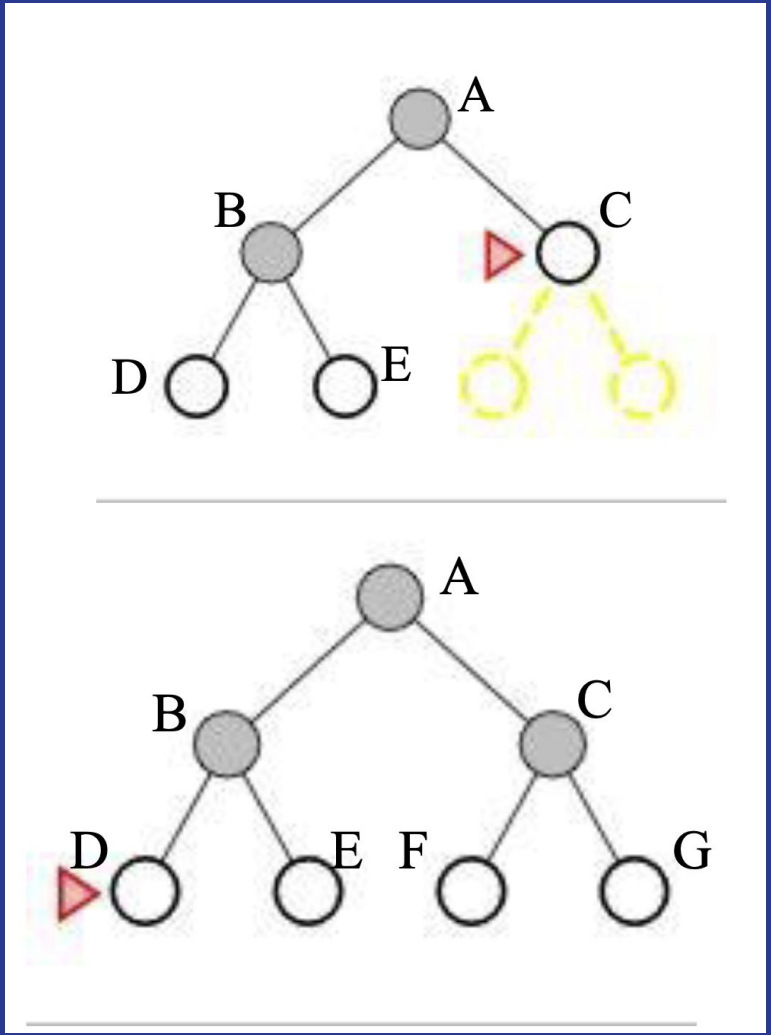
Cost Function: Number of assignments

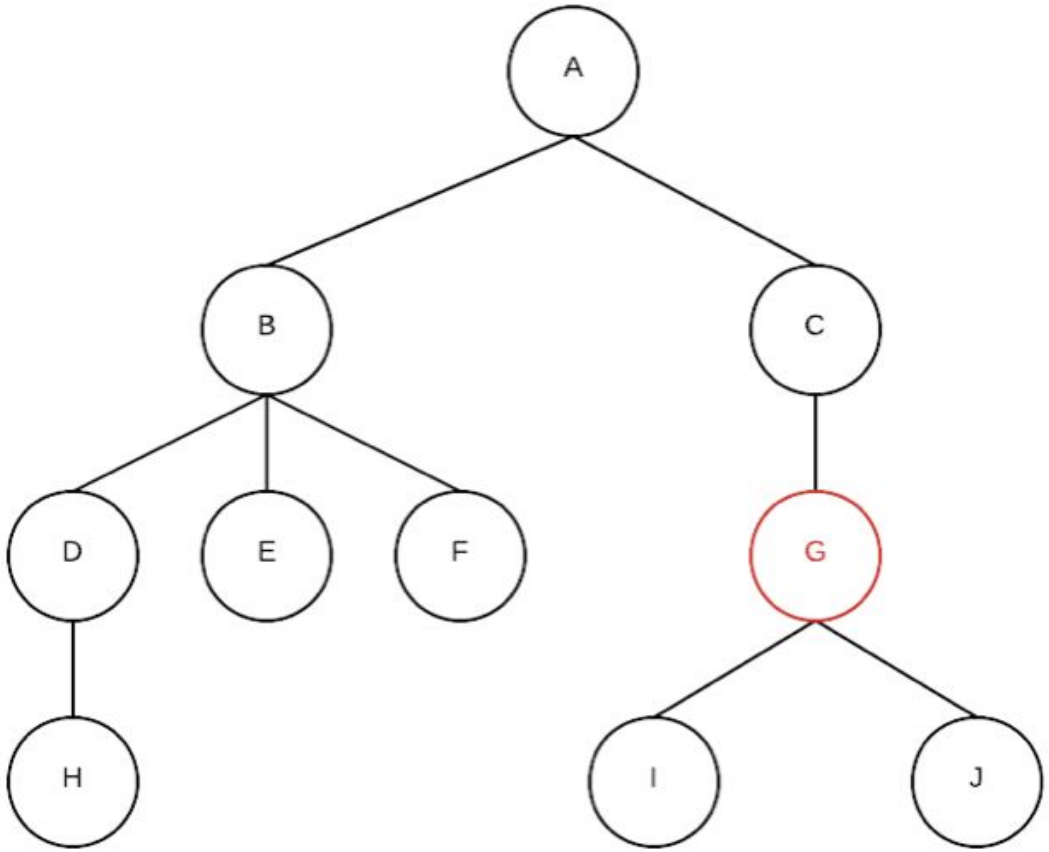
Exercise 5.1: Consider the following state-space diagram, where the lines are directed from top to the bottom, i.e. one can e.g. move from state A to B, but not back from B to A, etc.



BFS

FIFO




A
B
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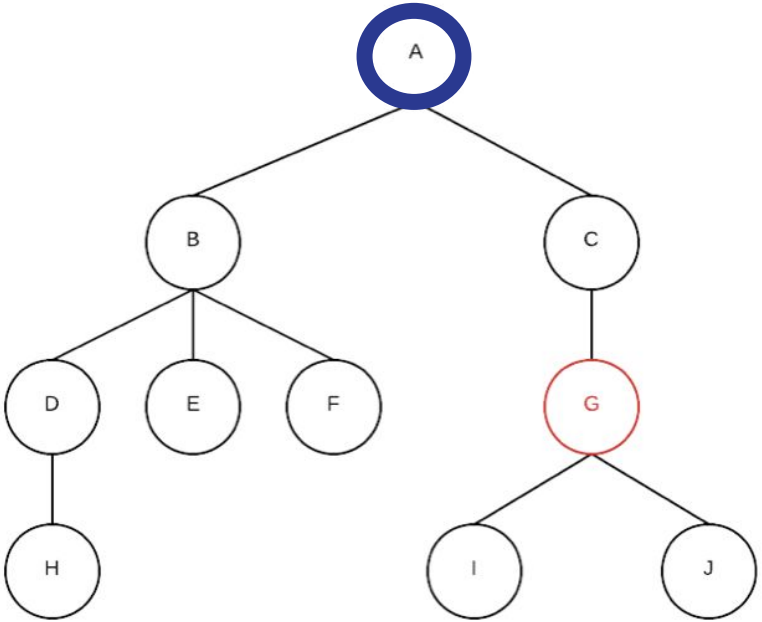
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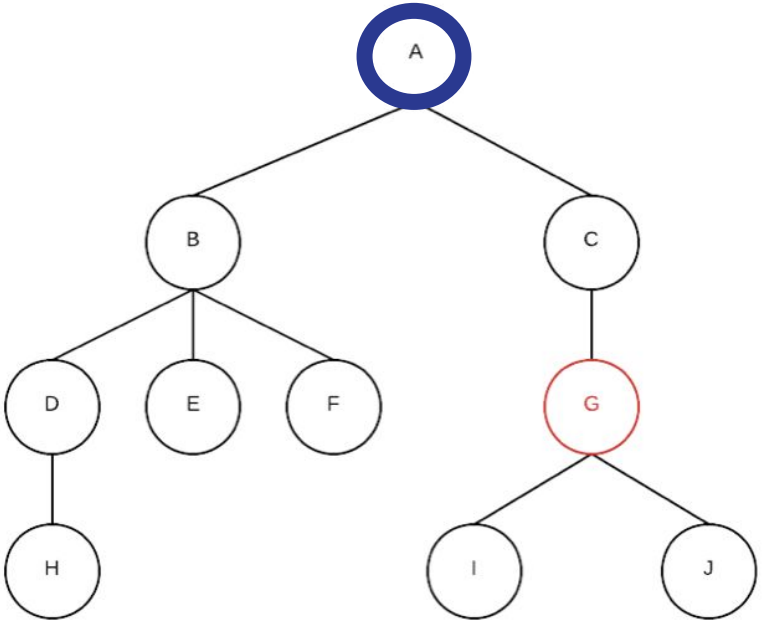
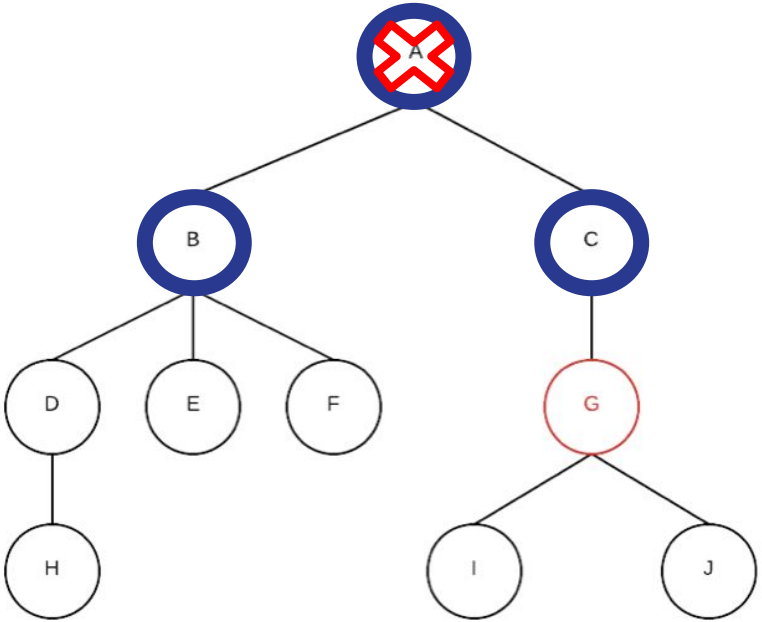
A



C

B

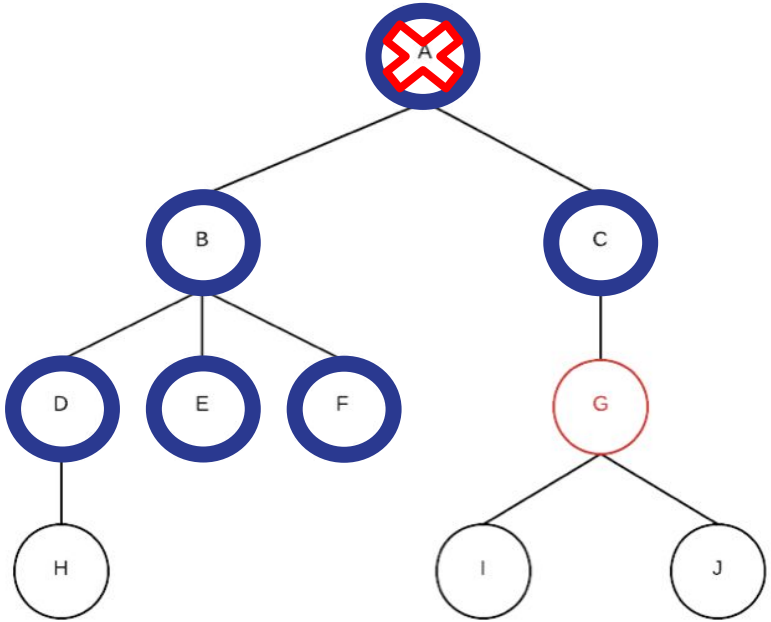
~~A~~



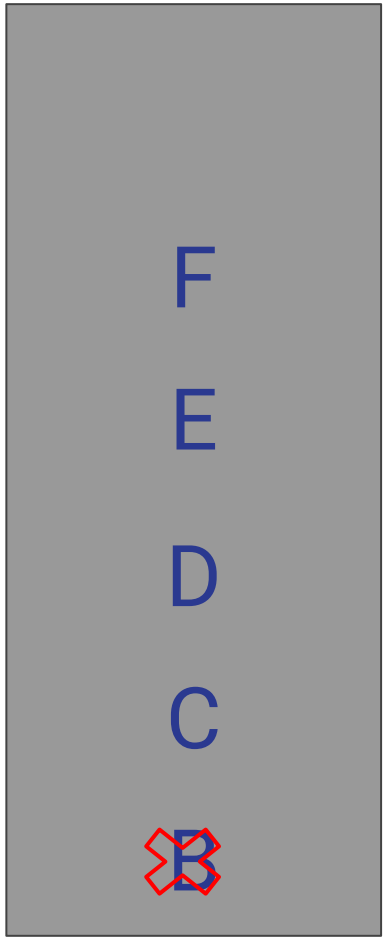
____ Result: A

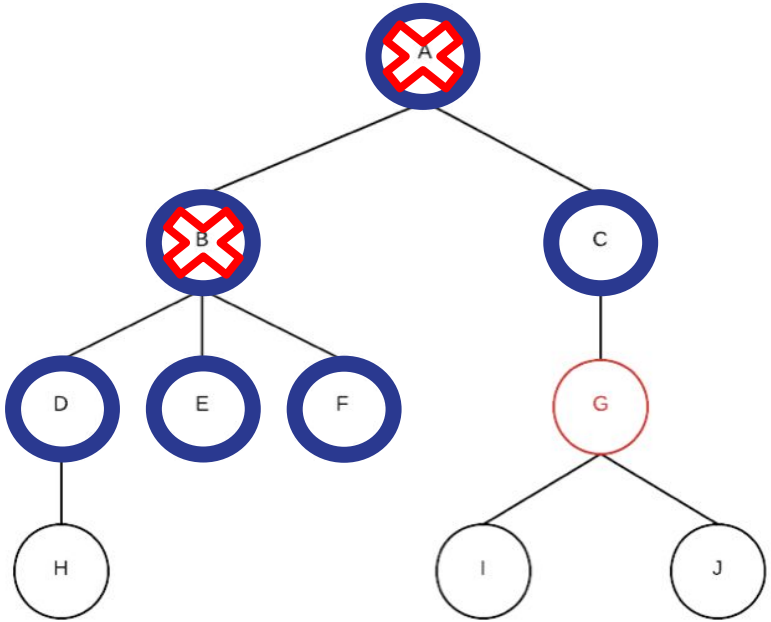
C

B



____ Result: A





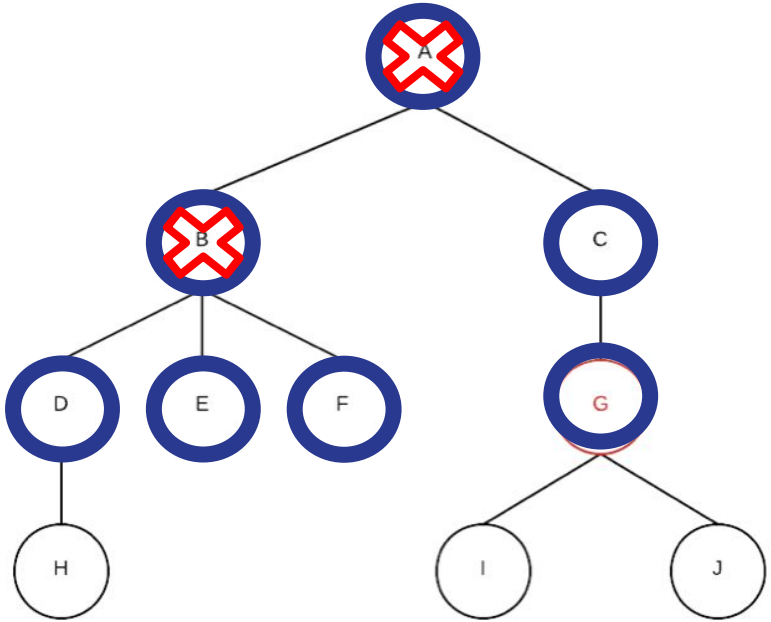
Result: A B

F

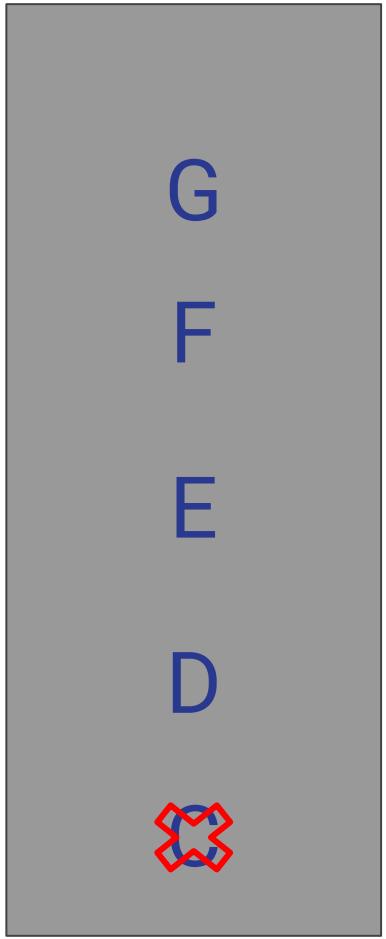
E

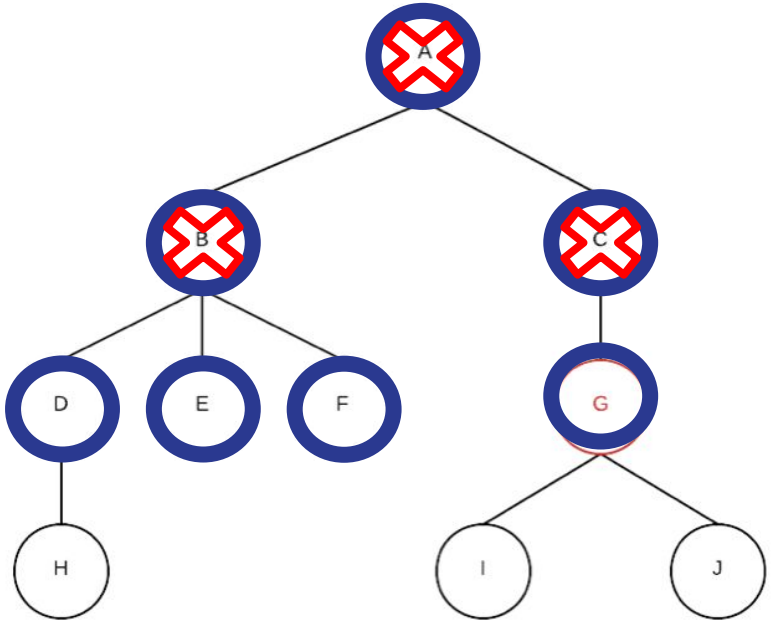
D

C



____ Result: A B





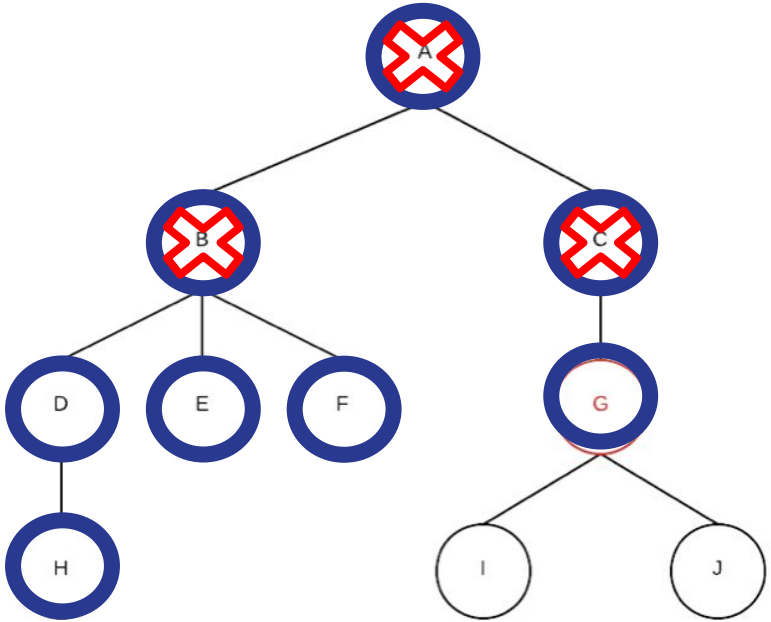
Result: A B C

G

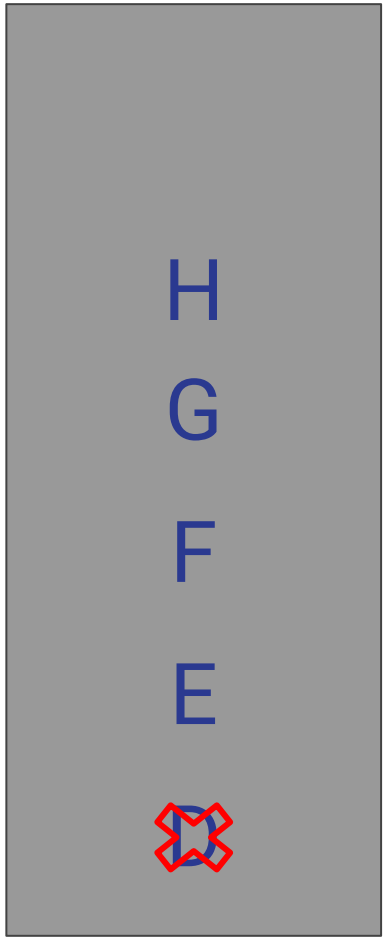
F

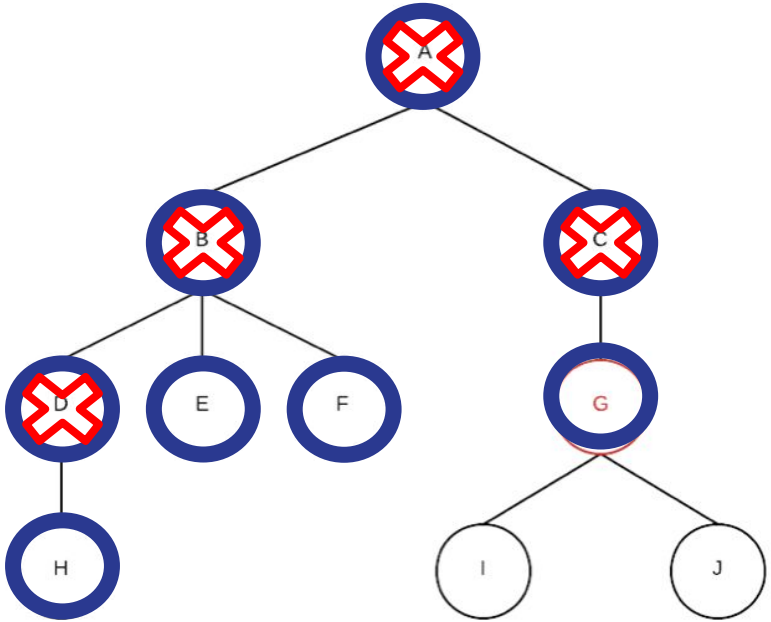
E

D

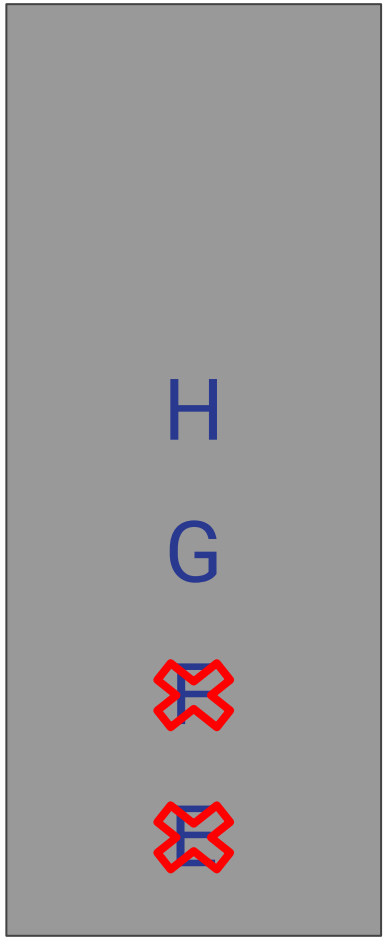


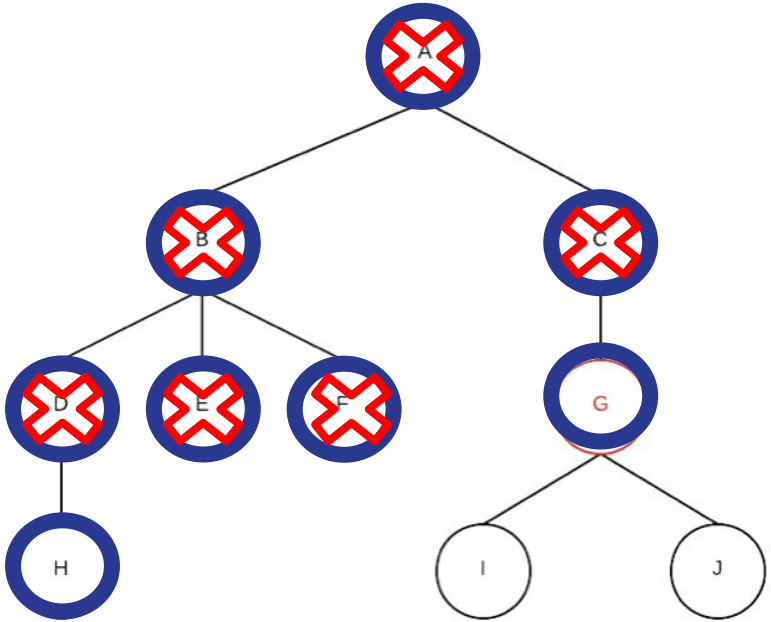
____ Result: A B C





Result: A B C D

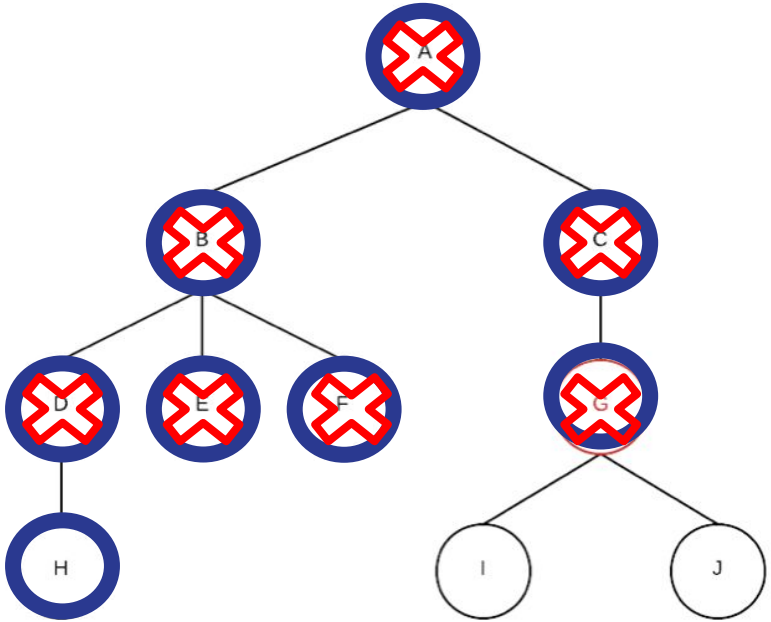




Result: A B C D E F

H

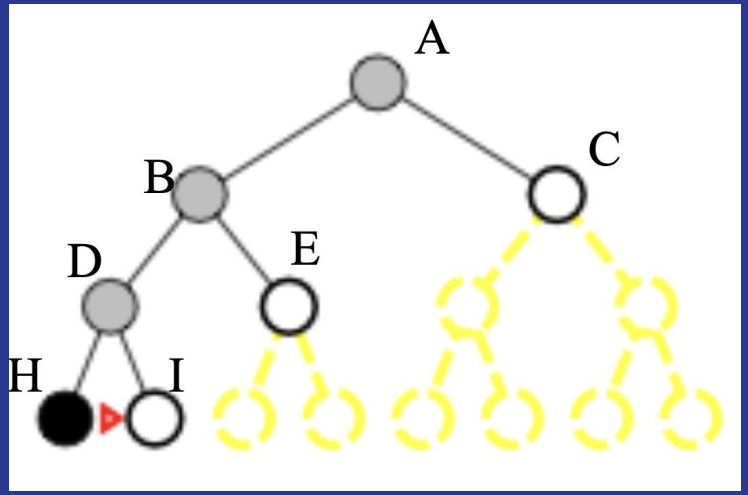
G

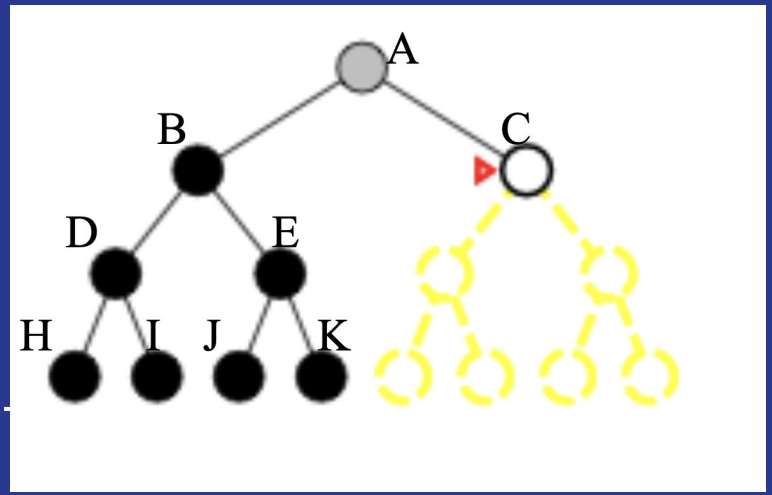


Result: A B C D E F G

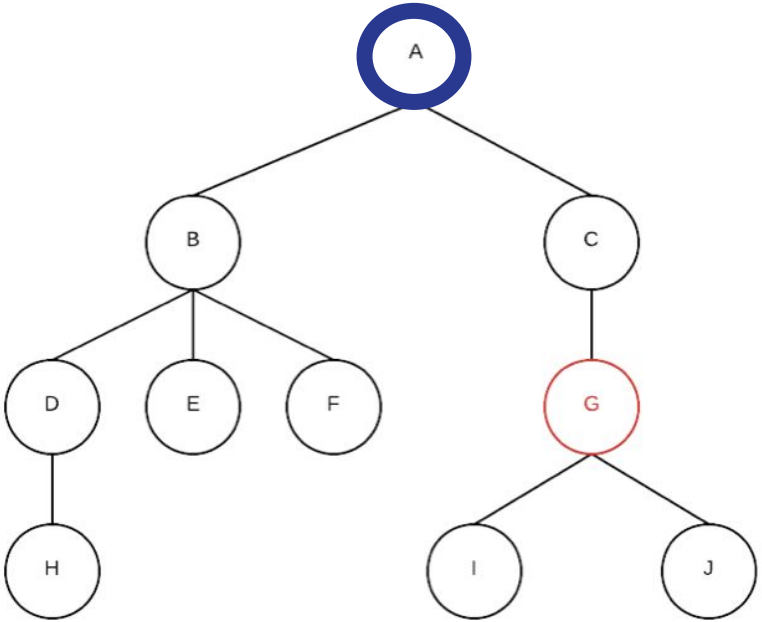
DFS

LIFO - queue



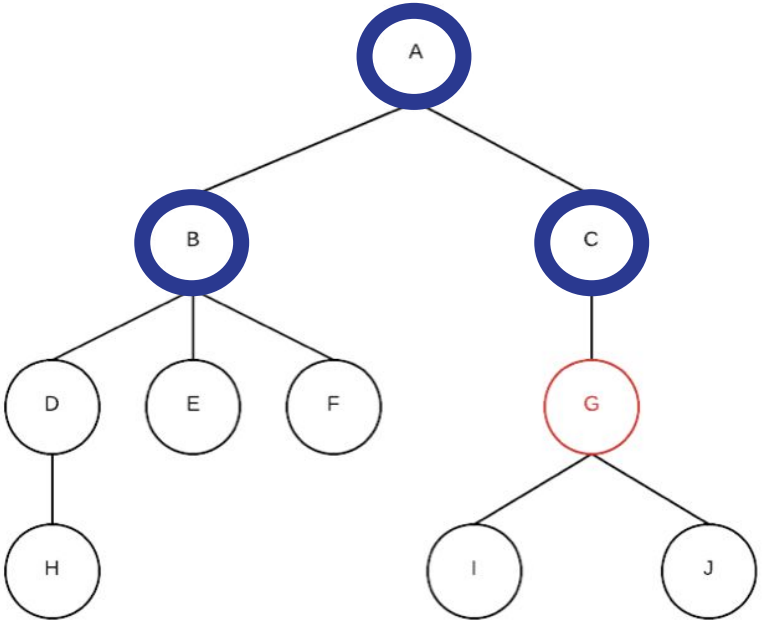


A



C

B



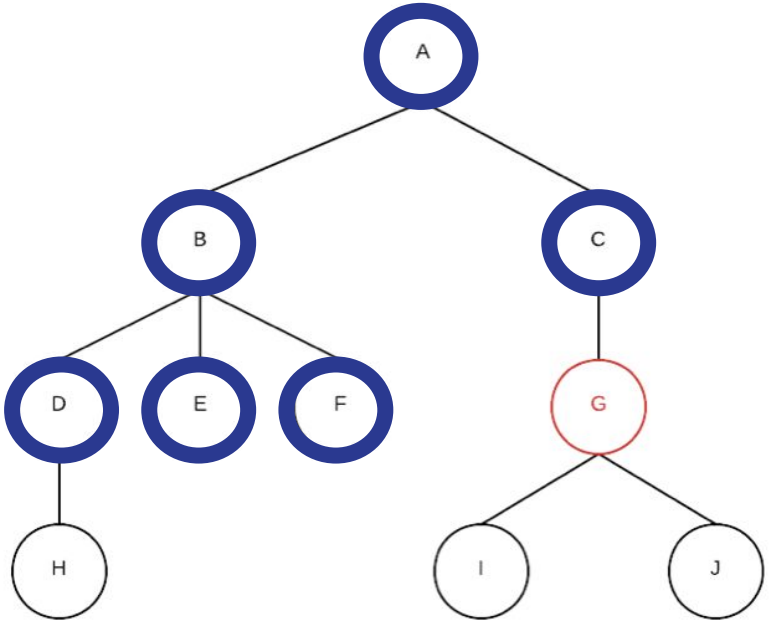
____ Result: A

C

F

E

D



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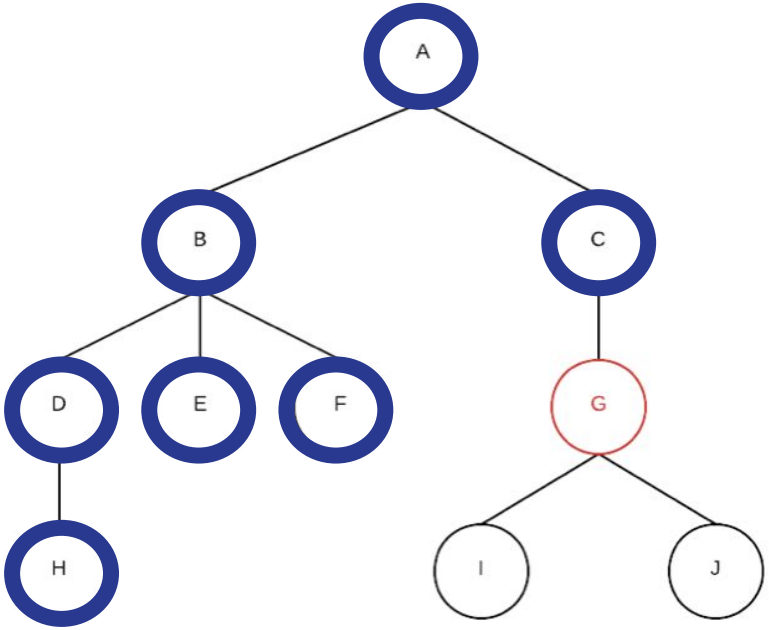
C

F

E

H

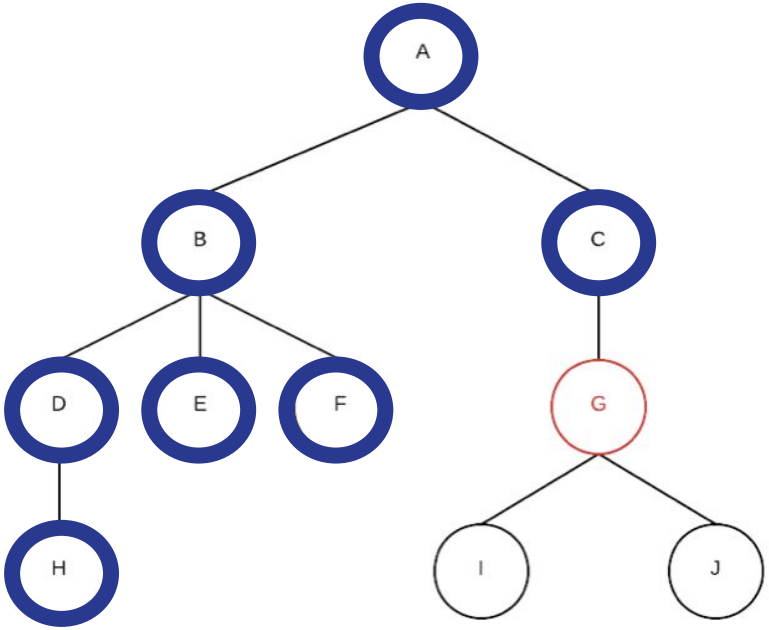


____ Result: A B D

C

F

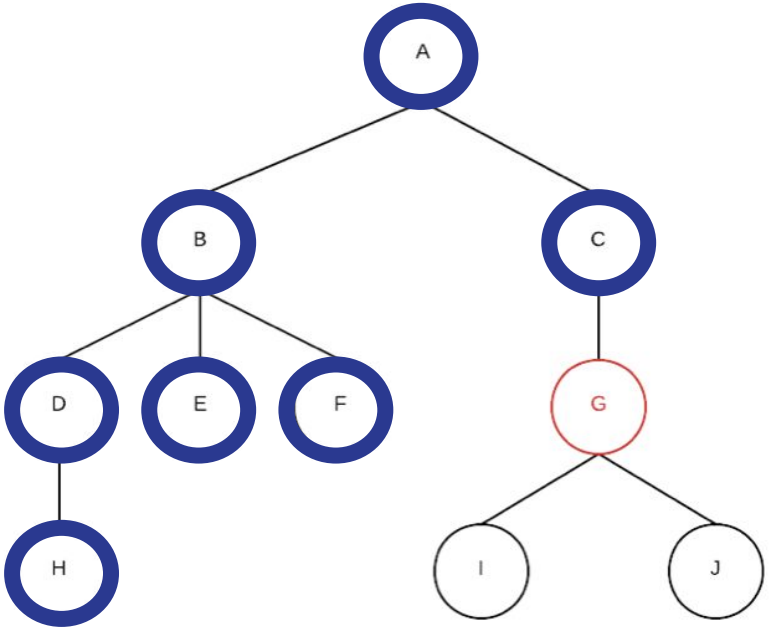
E



____ Result: A B D H

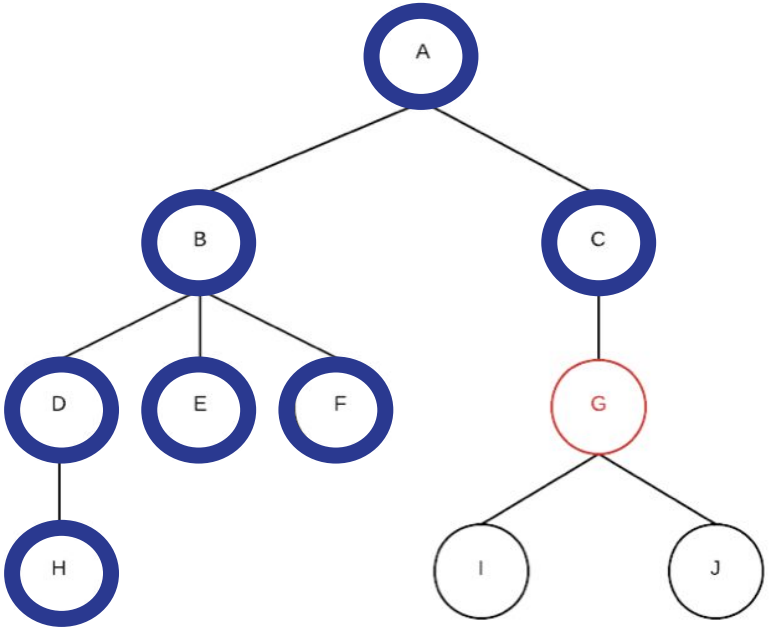
C

F



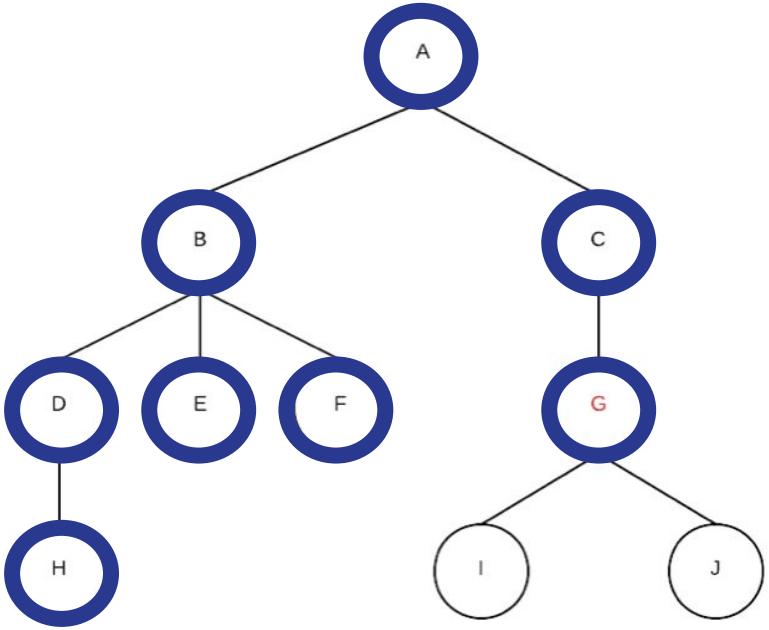
____ Result: A B D H E

C



____ Result: A B D H E F

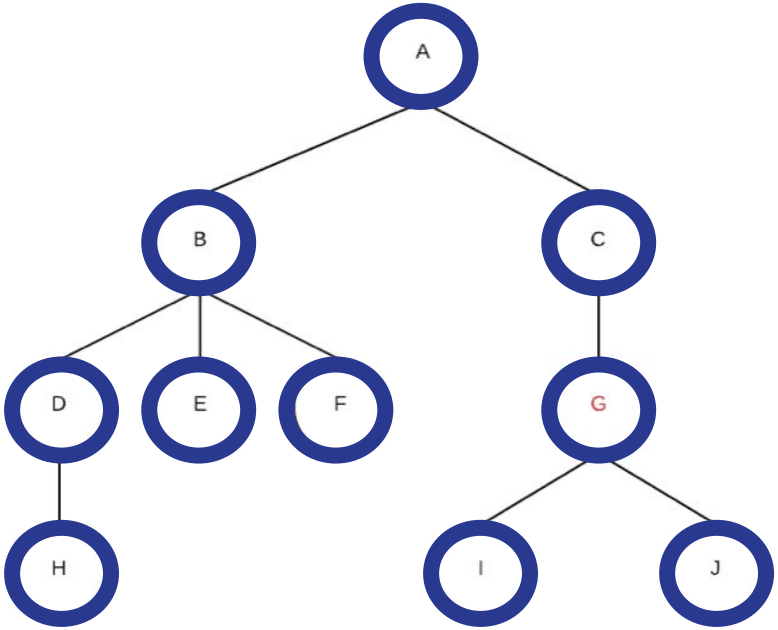
G



____ Result: A B D H E F C

J

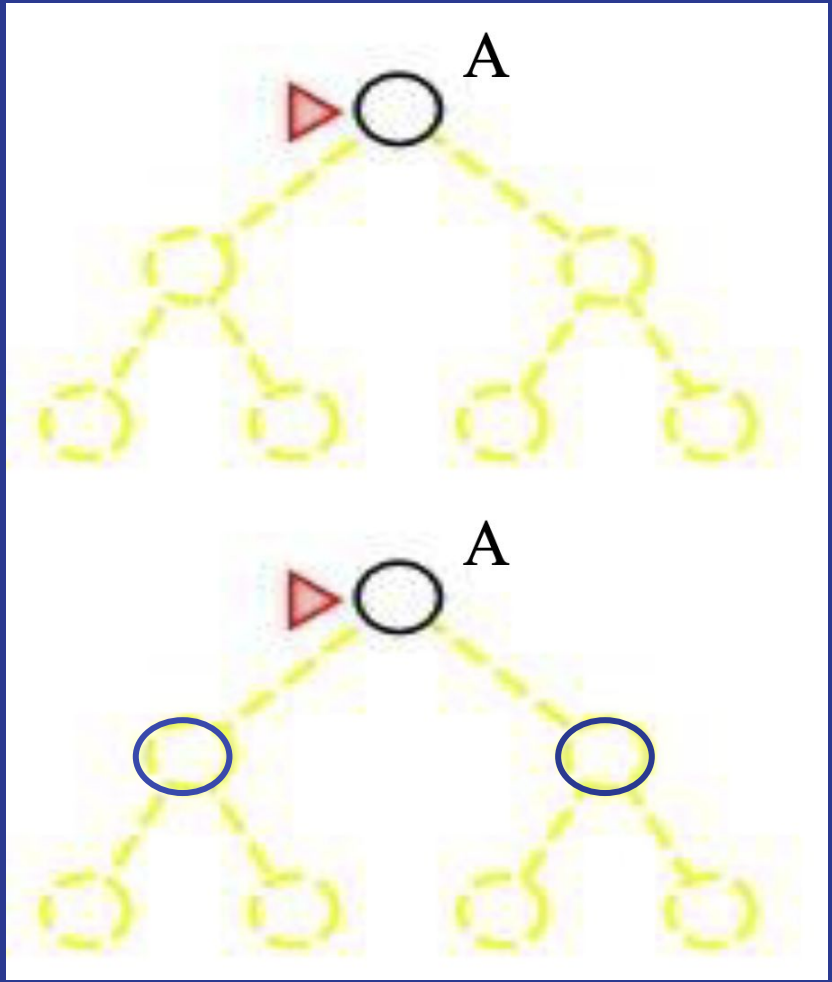
I



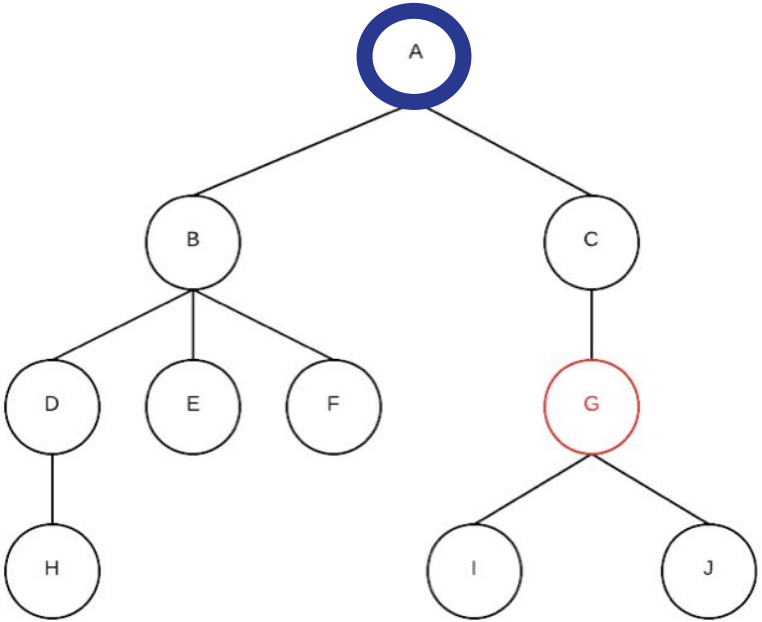
____ Result: A B D H E F C G

Depth-first iterative deepening

Per level



A

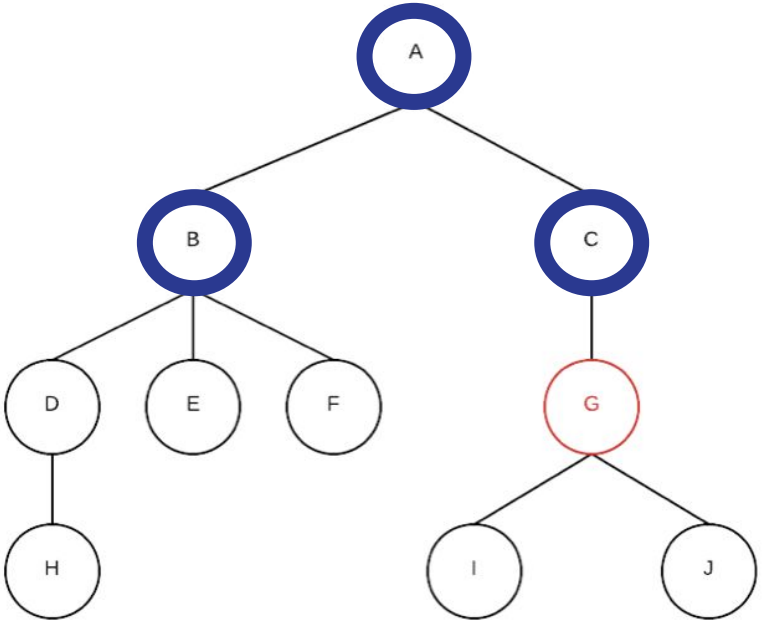


____ Result: A

C

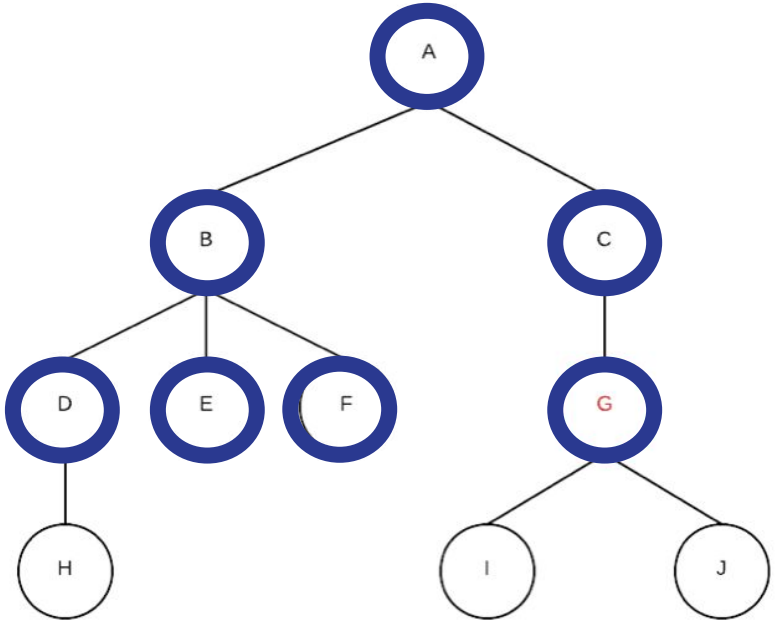
B

A



____ Result: A – A B C

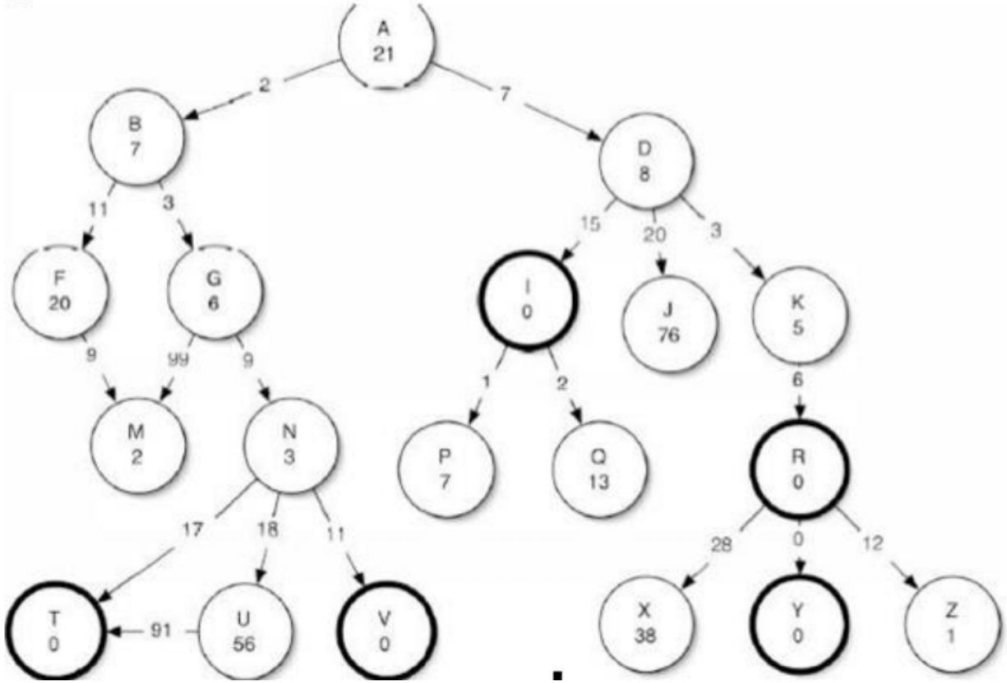
G
F
E
D
C
B
A



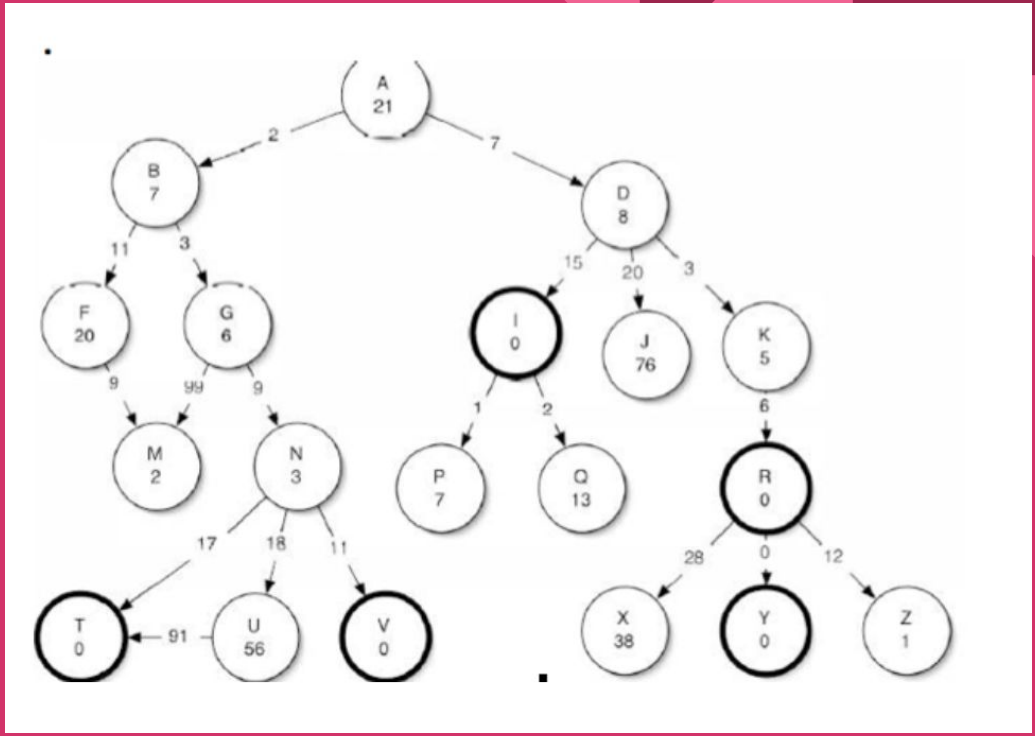
Result: A – A B C – A B C D E F G

Exercise 5.2: Consider the following state-space, where the goal states are marked with thicker boundaries.

a) When performing **depth-first search**, what is the order of the states that will be searched before you find the first goal state? Give the answer in letters.
b) Do the same for **breadth-first search**.
c) And for **depth-first iterative deepening**.



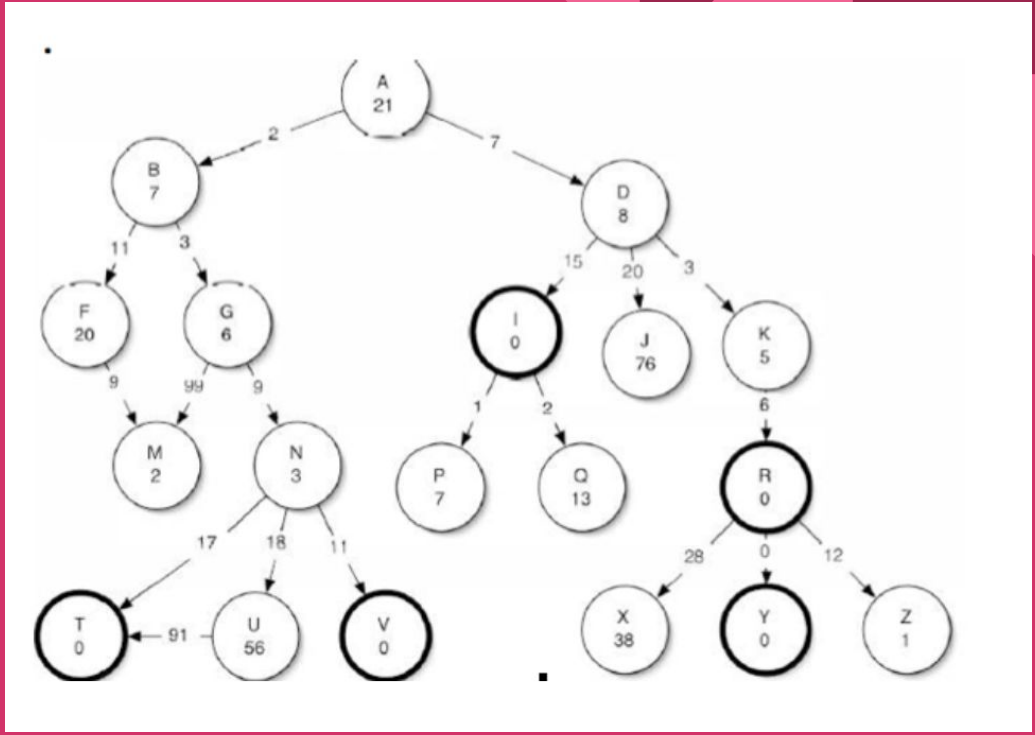
A) DFS
B) BFS
C) Depth first….



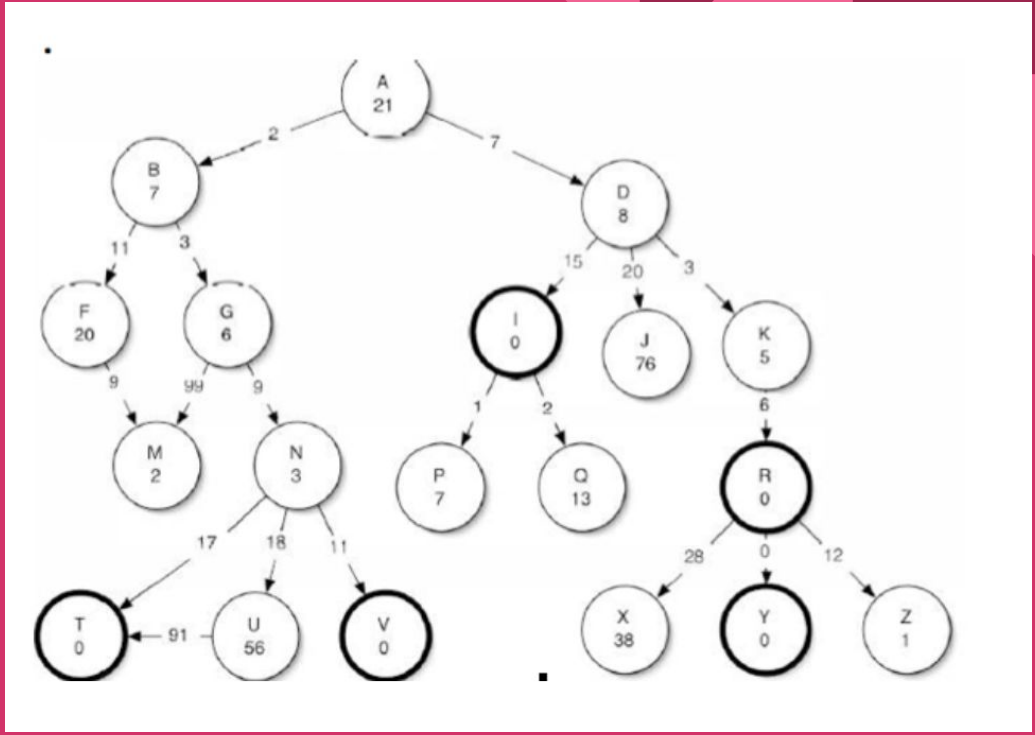
A) DFS

Respond:

A B F M G (M) N T

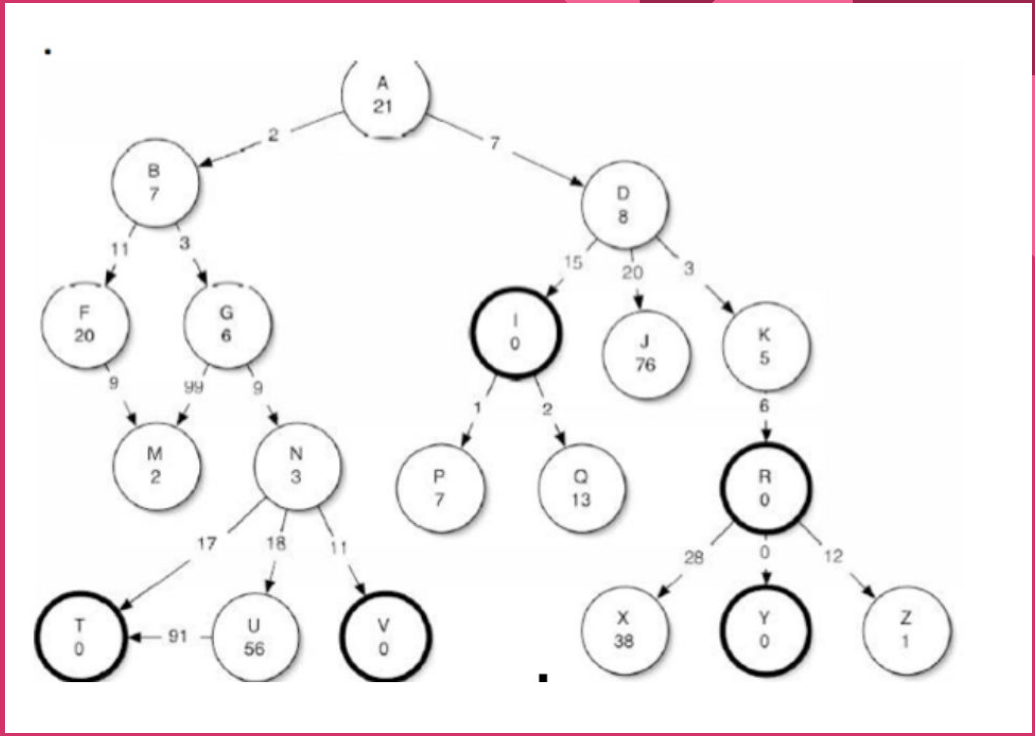


B) BFS

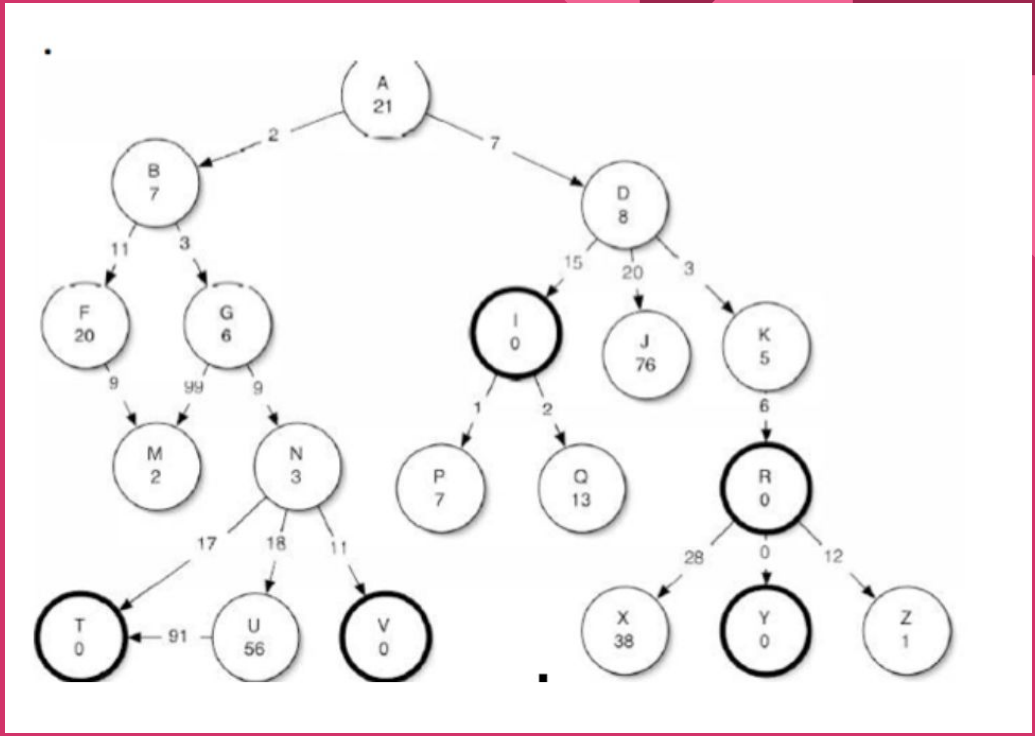

A
21
2
7
B
7
D
8
11
3
15
20
3
F
20
G
6
I
0
J
76
K
5
9
99
9
1
2
6
M
2
N
3
P
7
Q
13
R
0
17
18
11
28
0
12
T
0
91
U
56
V
0
X
38
Y
0
Z
1

B) BFS Respond:

A B D F G I



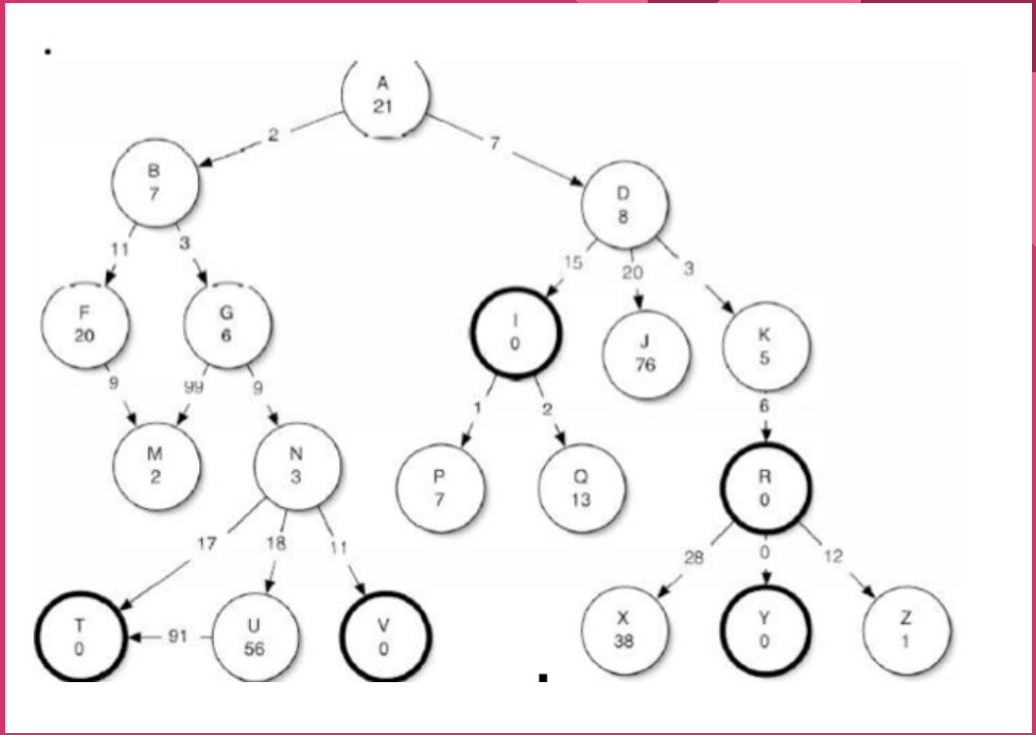
C) Depth First



C) Depth First interactive deeping

Respond:

A – ABD –ABDFGIJK



5.3 When do you advise to search **data-driven** instead of **goal- driven** ?

Data-driven when there are too many goal states, and in case goal states are not explicitly known.

You should also look at the branching factors in both directions. When this size is larger when starting from the Initial State you should, e.g., do goal directed search.

5.4 What are the advantages and disadvantages of breadth-first compared to depth-first searching algorithms? Discuss at least two different aspects.

5.4 What are the advantages and disadvantages of breadth-first compared to depth-first searching algorithms? Discuss at least two different aspects.

- Breadth-first search guarantees to find a solution
- the shortest path first
- Depth-first search can get stuck in a search space that is not finite
- Depth-first has advantages when the searching tree has a high branching factor + smaller space complexity

5.5 Give an example of a search tree in which depth-first iterative deepening would be much more expensive (in terms of computing time) to use compared to depth-first search?

5.5 Give an example of a search tree in which depth-first iterative deepening would be much more expensive (in terms of computing time) to use compared to depth-first search?

When the solution is deep in the left side